

M-Pro URV Dataset

Repository name: MPro-URV

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DATA DESCRIPTION

The database is composed of five folders and five files.

- Folder **Complex**. The subfolder **ALIGNED** contains 378 structures of the SARS-CoV-2 M-pro in complex with non-covalent inhibitors, including atomic coordinates derived from X-ray crystallographic structures. All protein–ligand complexes are structurally aligned based on the protein backbone and provided in a common structural reference frame to enable direct comparison across complexes. These proteins are presented in CIF format. For traceability purposes, the subfolder **CIF_FROM_PDB_NOT_ALIGNED** contains the corresponding unaligned complex structures in compressed CIF format (cif.gz) as retrieved directly from the Protein Data Bank (PDB). In addition, the subfolder **PROTONATED_NOT_ALIGNED** includes protonated, unaligned structures of the same complexes downloaded directly from the Protein Data Bank in Europe (PDBe). These protonated structures are required for the calculation of protein–ligand interactions using the Arpeggio program.
- Folder **Protein**. This folder contains the isolated protein components corresponding to the 378 SARS-CoV-2 main protease (M-pro) structures. The protein-only structures were extracted from the aligned protein–ligand complexes included in the **Complex/ALIGNED** subfolder. As a result, all protein structures are provided in a common structural reference frame, enabling direct comparison across the dataset. The folder is organized into three subfolders: **Protein_CIF**, which contains the protein structures in CIF format; **Protein_PDB**, which contains the same protein structures in PDB format; and **Protein_DSSP**, which contains DSSP files describing the secondary structure assignment and solvent accessibility of the protein residues derived from the aligned protein structures..
- Folder **Ligand**. This folder contains the 378 ligand structures corresponding to the non-covalent inhibitors of the SARS-CoV-2 main protease (M-pro). Each ligand was extracted from the aligned protein–ligand complexes in the **Complex/ALIGNED** subfolder, retaining only the inhibitor of interest. As a result, all ligand structures are provided in a common structural reference frame consistent with the aligned protein structures. Ligands are provided in SDF, CIF, and SMILES (SMI) formats. The SDF and CIF files contain three-dimensional atomic coordinates, whereas only the SDF format explicitly encodes bond orders between atoms. The SMILES representations include stereochemical information describing the isomeric state of the compounds. For consistency across the dataset, all files are named using the corresponding PDB identifier. Files include the suffix “_protein”, whereas files in the Ligand folder include the suffix “_ligand”. The three- or five-letter ligand identifiers are reported in the Info.csv file.

- Folder **Interaction**. It contains a list of 378 JSON files containing the interactions between the ligand and the M-pro. **Figure 1** illustrates an example of the interaction data stored in the JSON file between a ligand and the M-pro. The interacting entities field indicates that the interaction occurs between the ligand and the protein (INTER). In this example, the CD2 atom of residue 163, a Histidine (HIS) in chain A of the protein, forms an aromatic interaction with the N3 atom of the ligand SQ2. The distance between these interacting atoms is 3.9 Å.

```
{
  "bgn": {
    "auth_asym_id": "A_2",
    "auth_atom_id": "CD2",
    "auth_seq_id": 163,
    "label_comp_id": "HIS",
    "label_comp_type": "P",
    "pdbx_PDB_ins_code": " "
  },
  "contact": [
    "proximal",
    "aromatic"
  ],
  "distance": 3.9,
  "end": {
    "auth_asym_id": "A_2",
    "auth_atom_id": "N3",
    "auth_seq_id": 401,
    "label_comp_id": "SQ2",
    "label_comp_type": "B",
    "pdbx_PDB_ins_code": " "
  },
  "interacting_entities": "INTER",
  "type": "atom-atom"
},
```

Figure 1: Example of a non-covalent interaction between the ligand SQ2 and the protein in the 7B2J structure.

- Folder **Splits**. It contains three files in which there is the code of the complex used for testing, validation and training. There are five splits.
- File **pIC50.txt**. This text file contains two columns: the first one contains the PDB identifier of each complex, and the second one reports the corresponding pIC₅₀ value in decimal format.
- File **Info.csv**. This csv file provides additional metadata for each protein–ligand complex included in the dataset, including the PubMed ID and DOI of the publication describing the structure, ligand SMILES strings, ligand identifiers, Protein sequences and related structural information.
- File **Ligands_2D.pdf**. This file provides 2D representations of all 378 ligands included in the dataset.
- File **Data_Description.pdf**. This document.

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